STALKS OF THE DERIVED LOWER SHRIEK ARE COMPACTLY SUPPORTED COHOMOLOGY OF THE FIBRE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change for the derived lower shriek over a cartesian square of quasi-compact quasi-separated schemes with f separated finite type: g^{-1} composed with Rf_! equals Rf'_! composed with (g')^{-1} (more-etale, Lemma lemma-base-change-shriek, canonical tag 0F7L). cited at proof line 4324. The named input: pulling back Rf_!K to the geometric fibre over a point of Y is computed by Rf'_! on the fibre, which is the compactly supported cohomology construction of the fibre.

Source reference. Original lemma 0F7L: sources/stacks-chow/tags.txt, source lines 19011–19011. Cited at proof line 4324: the base change isomorphism computing the stalk of Rf_!K over a geometric point.

Lemma 0.2. *Let $f : X \to Y$ be a finite type separated morphism of schemes with Y quasi-compact and quasi-separated. Let K be an object of $D^+_{tors}(X_{étale}, \Lambda)$ or of $D(X_{étale}, \Lambda)$ in case Λ is torsion. Then there is a canonical isomorphism*

$$(Rf_!K)_{\overline{y}} \longrightarrow R\Gamma_c(X_{\overline{y}}, K|_{X_{\overline{y}}})$$

in $D(\Lambda)$ for any geometric point $\overline{y} : \mathrm{Spec}(k) \to Y$.

Proof. Immediate consequence of Lemma 0F7L and the definitions. □

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